

1. [5 pts] Find the general antiderivative of the following functions. Use C as an arbitrary constants.

1). $f(x) = 4 \sin x$

2). [3 pts] $f(x) = \frac{1}{x^4}$

2. [5 pts] Find the following indefinite integrals. Use C as an arbitrary constants.

1). $\int \frac{1}{1+x^2} dx$

2). $\int \frac{x^4 + x^2}{x^3} dx$